

Cumulative Distribution Functions and UPM/LPM Analysis

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Abstract

We show that the Cumulative Distribution Function (CDF) is represented by the ratio of the lower partial moment (LPM) ratio to the distribution for the interval in question. The addition of the upper partial moment (UPM) ratio enables us to create probability density functions (PDF) for any function without prior knowledge of its characteristics. We are able to replicate discrete distribution CDFs and PDFs for normal, uniform, poisson, and chi-square distributions, as well as true continuous distributions. This framework provides a new formulation for UPM/LPM portfolio analysis using co-partial moment matrices which are positive symmetrical semi-definite, aggregated to yield a positive symmetrical definite matrix.

I. Introduction:

The Empirical Cumulative Distribution Function (EDF) should, most of the time, be a good approximation of the true cumulative distribution function (CDF) as the sample set increases. This generalization is at the heart of statistics. Means and variances are used to assign and fit a distribution, but partial moments stabilize with a smaller sample size ensuring a more accurate analysis of the EDF.

The empirical CDF is a simple construct. It is simply the number of observations less than or equal to a target, divided by the total number of observations in a given data set. The problem with extrapolating these results to an assumed true CDF is that the discrete empirical CDF is extremely sensitive to sample size,¹ and any parameter nonstationarity will deteriorate the fit to the true distribution. The paper is organized as follows:

First, we propose a method to derive the CDF and PDF of the EDF, utilizing the upper and lower partial moments (*UPM* and *LPM* respectively) of the EDF. The benefits are obvious, such as compensating for any observed skewness and kurtosis that would force a more esoteric distribution family onto the data. These measurements require zero knowledge of the underlying function and no goodness-of-fit tests to approximate a likely true distribution. Partial moments also happen to exhibit less sample size sensitivity than means and variances as we will discuss later.

Next, this foundation is then used to develop conditional probabilities and joint distribution co-partial moments. Finally, this toolbox allows us to propose a new formulation for UPM/LPM analysis and we note that each of the co-partial moment matrices are positive symmetrical semi-definite, ensuring a positive symmetrical definite aggregate matrix. This represents a major

¹ Estimated mean average deviations are provided in Appendix A.

improvement in the use of partial moment matrices in portfolio theory and avoids the problems with co-semivariance matrices as noted by Grootveld and Hallerbach (1999) and Estrada(2008).

II. Deriving Cumulative Distribution and Partial Density Functions Using Partial Moments

A distribution may be dissected into two partial moment segments using an arbitrary target as shown in Figure 1.

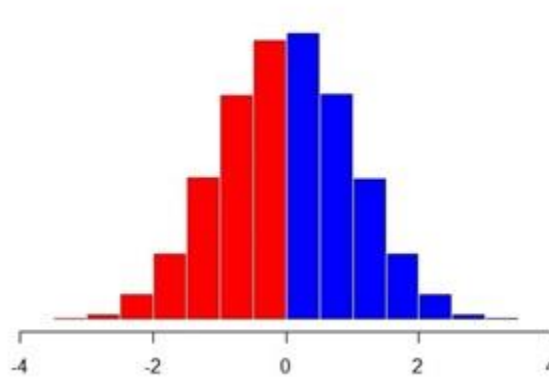


Figure 1. A distribution dissected into its two partial moment segments, red LPM and blue UPM, from a shared target.

The Upper and Lower partial moment formulas are below in Equations 1 and 2:

$$LPM(n, h, x) = \frac{1}{T} \left[\sum_{t=1}^T \max\{0, h - x_t\}^n \right] \quad (1)$$

$$UPM(q, l, x) = \frac{1}{T} \left[\sum_{t=1}^T \max\{0, x_t - l\}^q \right] \quad (2)$$

where x_t represents the observation x at time t , n is the degree of the LPM, q is the degree of the UPM, h is the target for computing below target returns, and l is the target for computing above target returns.²

One can visualize how the entire distribution is quantified with the upper and lower partial moment from the same target, ($h = l = 0$) in Figure 1. The area under the function derived from degree one partial moments will approximate the area derived from the integral of the function over an interval $[a, b]$ asymptotically. This asymptotic relationship is demonstrated in Viole and Nawrocki (2012) and represented with equation (3).

$$\lim_{t \rightarrow \infty} [UPM(1, 0, f(x)) - LPM(1, 0, f(x))] = \frac{\int_a^b f(x) dx}{(b - a)} \quad (3)$$

We use a degree zero ($n=q=0$) to generate a discrete analysis, replicating results from the conventional CDF and PDF methodology. Degree one ($n=q=1$) is used to generate the continuous results. This is an important distinction, as the discrete analysis is a relative frequency and probability investigation; while the continuous analysis integrates a variance consideration to capture the rectangles of infinitesimal width in deriving an area under a function. Standard deviation remains stable as sample size range increases, thus it is not an

² Equations 1 and 2 will generate a 0 for degree 0 instances of 0 results.

accurate barometer of the area of the function to estimate a continuous variable. Figure 2 illustrates the range increase as the number of observations increase for a normal distribution with $\mu=10$ and $\sigma=20$ for 5 million random draws from a normal distribution.

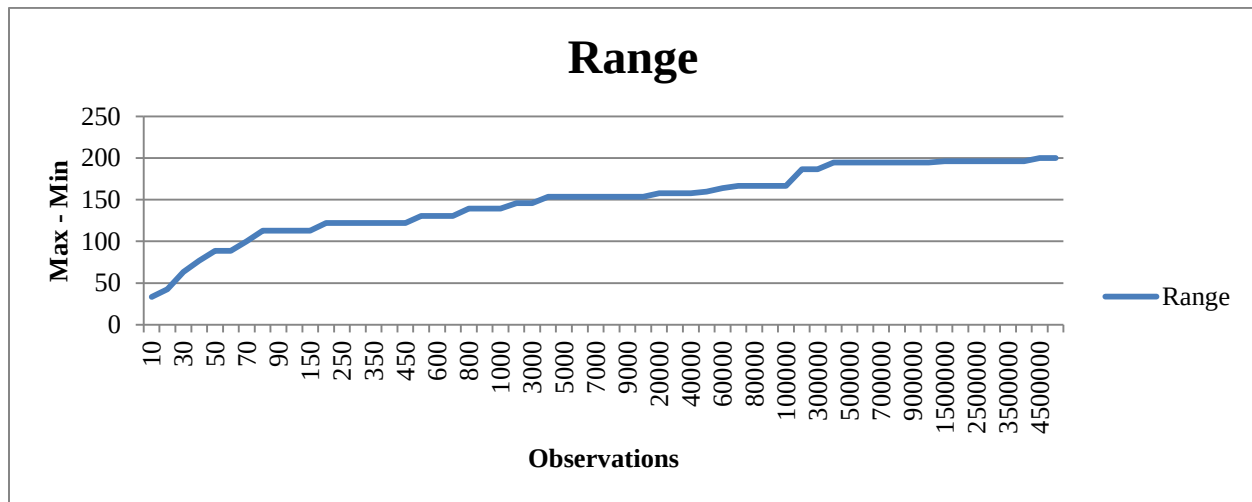


Figure 2. Range for a randomly generated normal distribution $\mu=10$ and $\sigma=20$ for 5 million random draws.

Just as the probability of two mutually exclusive events equal one, the sum of the ratios - LPM to the entire distribution; and UPM to the entire distribution (LPM_{ratio} and UPM_{ratio} respectively) plus the point probability, equal one as in equations 8 and 8a. The point probability is often included in the CDF calculation but it is not uniformly treated as less than or equal to the target.³

Theorem 1,

$$P\{X < x\} + P\{X > x\} + P\{X = x\} = 1 \quad (4)$$

³ There is no consensus language for CDF definitions. Some instances are “ $< x$ ” while others reference “ $\leq x$ ” depending on the distribution, discrete or continuous. We are uniform in our treatment of distributions with “ $\leq x$ ” for both discrete and continuous distributions. See <http://www.mathworks.com/help/toolbox/stats/unifcdf.html> and <http://www.mathworks.com/help/toolbox/stats/unidcdf.html> for treatment of the target, x .

If,

$$P\{X \leq x\} = LPM_{ratio}(0, x, X) = \frac{LPM(0, x, X)}{[LPM(0, x, X) + UPM(0, x, X)]} - \frac{\varepsilon}{2} \quad (5)^4$$

$$LPM_{ratio}(0, x, X) = LPM(0, x, X) \quad (5a)$$

$$LPM_{ratio}(1, x, X) \neq LPM(1, x, X) \quad (5b)$$

And,

$$P\{X > x\} = UPM_{ratio}(0, x, X) = \frac{UPM(0, x, X)}{[LPM(0, x, X) + UPM(0, x, X)]} - \frac{\varepsilon}{2} \quad (6)$$

$$UPM_{ratio}(0, x, X) = UPM(0, x, X) \quad (6a)$$

$$UPM_{ratio}(1, x, X) \neq UPM(1, x, X) \quad (6b)$$

Since the entire normalized distribution is represented by,

$$\left[\frac{LPM(0, x, X)}{[LPM(0, x, X) + UPM(0, x, X)]} - \frac{\varepsilon}{2} \right] + \left[\frac{UPM(0, x, X)}{[LPM(0, x, X) + UPM(0, x, X)]} - \frac{\varepsilon}{2} \right] + \varepsilon = 1 \quad (7)$$

Where ε is the point probability $P\{X = x\}$. The use of an empty set for ε yields,

$$LPM(0, x, X) + UPM(0, x, X) = 1 \quad (8)$$

⁴ It is important to note that $LPM(0, x, X)$ is a probability measure and will yield a result from 0 to 1. Thus, the ratio of $LPM(0, x, X)$ to the entire distribution ($LPM_{ratio}(0, x, X)$) is equal to the probability measure itself, $LPM(0, x, X)$.

$$LPM_{ratio}(1, x, X) + UPM_{ratio}(1, x, X) = 1 \quad (8a)$$

The point probability ε for a discrete distribution can easily be computed by the frequency of the specific point divided by the total number of observations. The point probability would be more relevant in a discrete distribution of integers, and has an inverse relationship to the degree of specification of the underlying variable. As the specification approaches infinity, ε approaches zero.

We know from calculus that the $\int_a^b(x)dx = F(b) - F(a)$ and if $F(b) = F(a)$, the integral of a point equals zero. Thus for a continuous distribution, there is no difference between $P\{X < x\}$ and $P\{X \leq x\}$ since $\varepsilon = 0$. If one wishes to subscribe to the notion that the sum of an infinite amount of points each equal to zero must sum to one per the integral definition, then equation 7 is simply reduced to equation 8a for continuous variables. However, equation 7 with degree 1 can also be used for the continuous variable to compensate for $\varepsilon > 0$ and generate a normalized continuous probability.

A. Review of the Literature

Guthoff et al (1997) illustrate how the value at risk of an investment is equivalent to the degree zero LPM. We confirm this derivation as the degree zero LPM does indeed provide a normalized solution. However, critical errors were made by Guthoff and in subsequent works by Shadwick and Keating (2002), and Kaplan and Knowles (2004).

The omega ratio is defined as,

$$\Omega(\tau) = \frac{\int_{\tau}^{\infty} [1 - F(R)] dR}{\int_{-\infty}^{\tau} F(R) dR} \quad (9)$$

Where $F(\cdot)$ is the CDF for total returns on an investment and τ is the threshold return. Guthoff and Shadwick and Keating's error was the use of degree one LPM (area) on a degree 0 LPM, the probability CDF of the distribution. Degree one LPM does not need to be performed on the probability CDF as they present.

The Kappa measure is defined as,

$$K_n(\tau) = \frac{\mu - \tau}{\sqrt[n]{LPM_n(\tau)}} \quad (10)$$

Kaplan and Knowles' error was the dismissal of the degree zero LPM (0-th root of something does not exist) which we show equals historical CDF measurements for various distributions. Also, $\sqrt[n]{LPM_n(\tau)}$ forces concavity upon increased n , which do not presume such a condition.

The omega ratio (Shadwick and Keating, 2002) and kappa measure (Kaplan and Knowles, 2004) both demonstrate the need for a full derivation of partial moments and their CDF equivalence with full degree explanation and relevance.

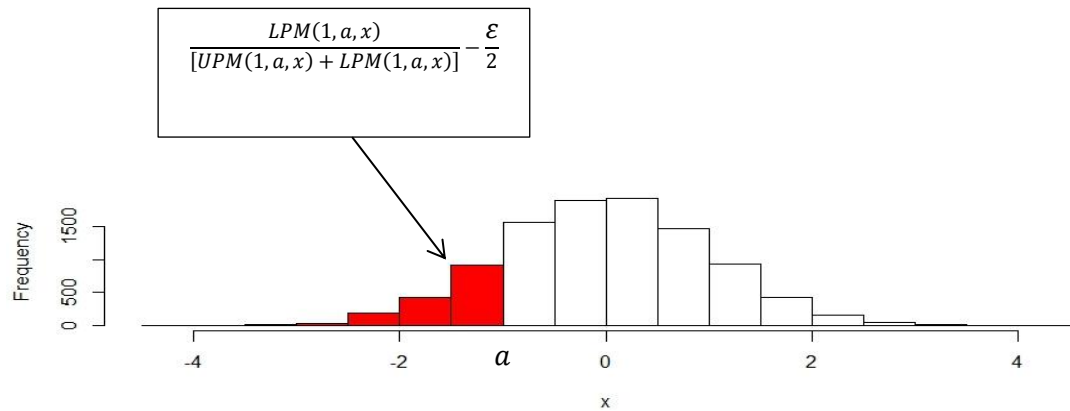


Figure 3. Area of a Probability Density Function represented by the Cumulative Distribution Function of an arbitrary point a for the interval $[-\infty, a]$.

Cumulative Distribution Function (CDF) using partial moments:

$$F_X(x) = P(X \leq x) \quad (11)$$

$$F(x) = \int_{-\infty}^x f(x)dx. \quad (12)$$

Discrete,

$$F(x) = LPM(0, a, x) \quad (13)$$

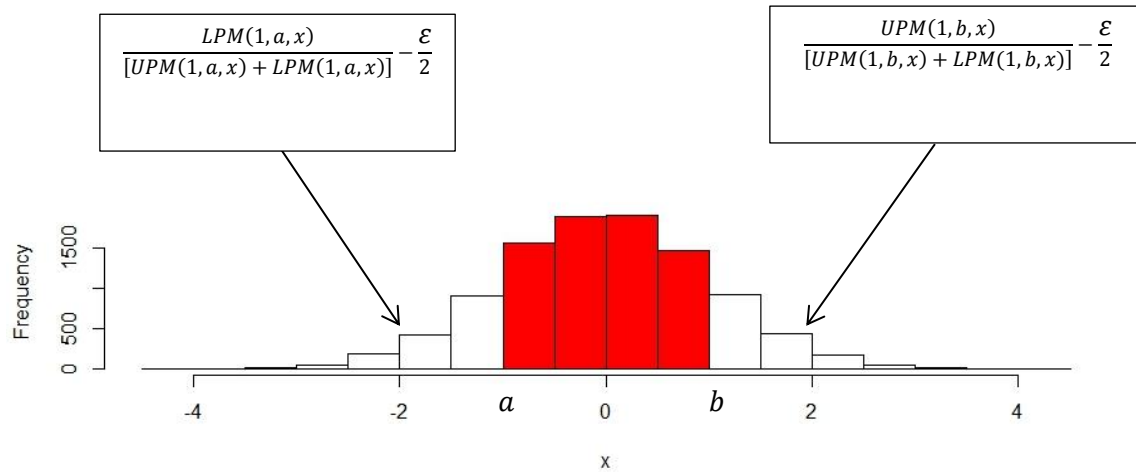
Continuous,

$$F(x) = LPM_{ratio}(1, a, x) \quad (14)$$

For any distribution the continuous estimate yields,⁵

$$0.5 = LPM_{ratio}(1, \mu, x) \quad (15)$$

⁵ Figure 7 offers a visual representation of the difference between continuous and discrete CDFs of the mean.



Probability Density Function (PDF) using partial moments:

$$P[a \leq x \leq b] = \int_a^b f(x) dx \quad (16)$$

Discrete,

$$P[a \leq x \leq b] = LPM(0, b, x) - LPM(0, a, x) \quad (17a)$$

Continuous,

$$P[a \leq x \leq b] = LPM_{ratio}(1, b, x) - LPM_{ratio}(1, a, x) \quad (17b)$$

B. Methodology Notes:

We generated random distributions for 5 million observations. We then took 300 iterations with different seeds and averaged them. For stability estimates, we generated mean average deviations (MAD) for each statistic over the 300 iterations for observations 30 through 5 million.

The statistics used in the following discussion are as follows: *CHIDF(target)* - Cumulative distribution function for the Chi-square distribution and specified target; *Kurtosis* - Relative Kurtosis measure of the entire sample; *Mean* - μ of the entire sample; *Norm Prob(target)* - Cumulative distribution function for the Normal distribution and specified target; *POIDF(target)* - Cumulative distribution function for the Poisson distribution and specified target; *Range* - Max observation – min observation for the entire sample; *SemiDev* - Semi-deviation of the sample using mean as the target; *Skew* - Skewness measure of the entire sample; *StdDev* - Standard deviation of the sample; *UNDF(target)* - Cumulative distribution function for the Uniform distribution.

All of the above mentioned distributions and targets can be easily verified by the reader with statistical software such as the ISML subroutine library. Furthermore, the direct computation of the partial moments can also be easily implemented into such software. The sample parameters generated were as follows:

Normal Distribution: $\mu = 10.00018$ $\sigma = 19.99976$

Poisson Distribution: $\theta = 9.999914$

Uniform Distribution $\mu = 10.00045$

Chi-Square Distribution: $v = 1$ $\mu = 0.999947$

C. Normal Distribution

We compare our metric to the traditional CDF, Φ , of a standard normal random variable.

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt$$

The probability generated from the normal distribution converges to $LPM(0,0,X)$ in approximately 90 observations as shown in Figure 5. $LPM(0,0,X)$ stabilizes with less observations than the normal probability (exhibiting a lower MAD) as shown in Appendix A, Table 1a. This is proof that $LPM(0,0,X)$ is indeed the discrete CDF of the distribution for the area less than the target. While the normal probability is less than or equal to the target compared to less than for $LPM(0,0,X)$; the probability of the specific target outcome does not affect the probability to the specification of four decimal places.

The relationship between $LPM_{ratio}(1,0,X)$, $LPM(0,0,X)$ and the normal probability or $Norm Prob(0)$ is shown in Figure 5. The further from the mean, the greater the discrepancy between the continuous and discrete CDF as seen in Figure 6. As the area of the distribution increases for the UPM if the target is less than the mean, the continuous CDF will be consistently lower than the discrete CDF. Conversely, as the area of the LPM increases if the target is greater than the mean, the continuous CDF will be consistently higher than the discrete CDF. This holds for all distribution families. The continuous and discrete probabilities are obviously equal at the endpoints of the distribution, 0 and 1 for minimum and maximum respectively.

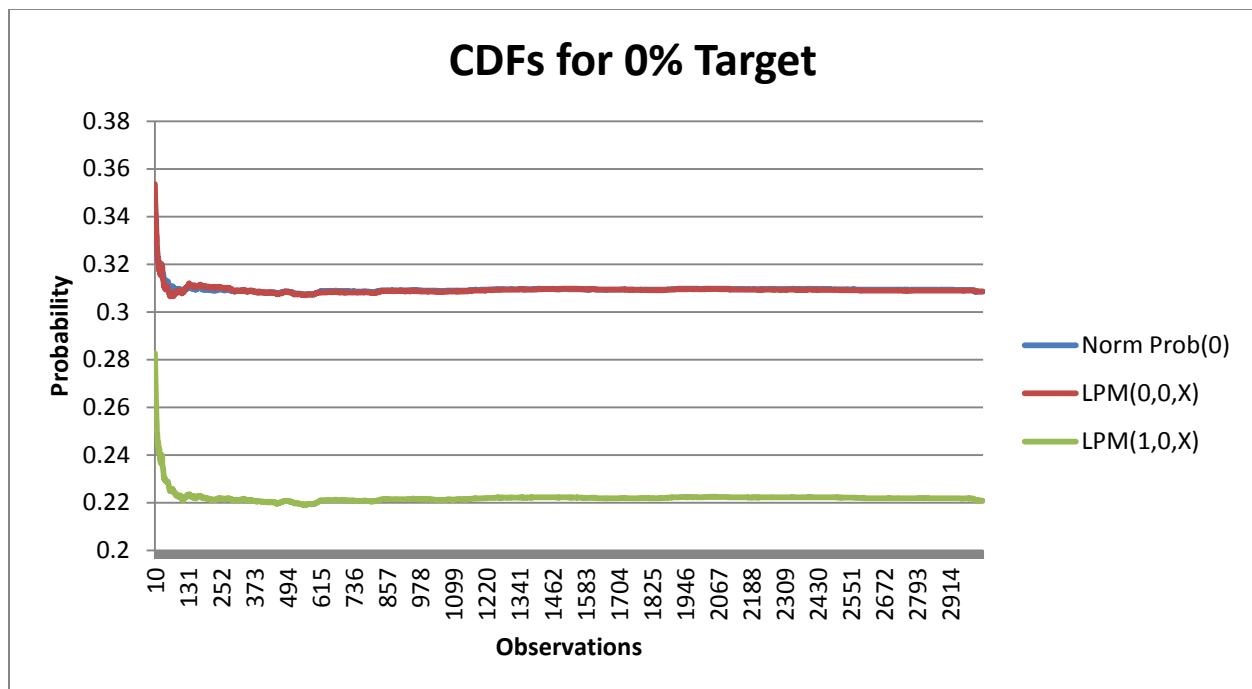


Figure 5. CDF of 0% target for Normal distribution with $\mu=10$ and $\sigma=20$ parameter constraints.

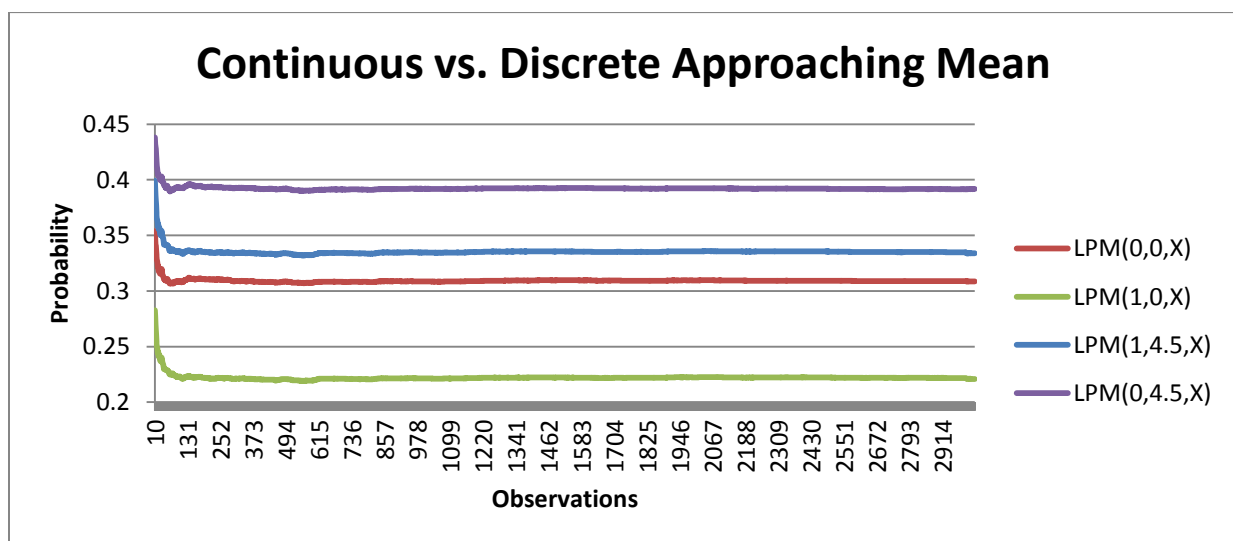


Figure 6. Continuous estimate converges towards discrete estimate as the target approaches sample mean (as h is increased from 0 to 4.5). The LPM $n=0$, $h=0$ is denoted as LPM(0,0,X), LPM $n=1$, $h=0$ is denoted by LPM(1,0,X), LPM $n=1$, $h=4.5$ is denoted as LPM(1,4.5,X) and the LPM $n=0$, $h=4.5$ is LPM(0,4.5,X).

In Figure 7, the plot shows the convergence of the discrete LPM degree 0 from the mean to the continuous LPM degree 1 using the mean as the target return. The discrete isn't stable until around 1000 observations.

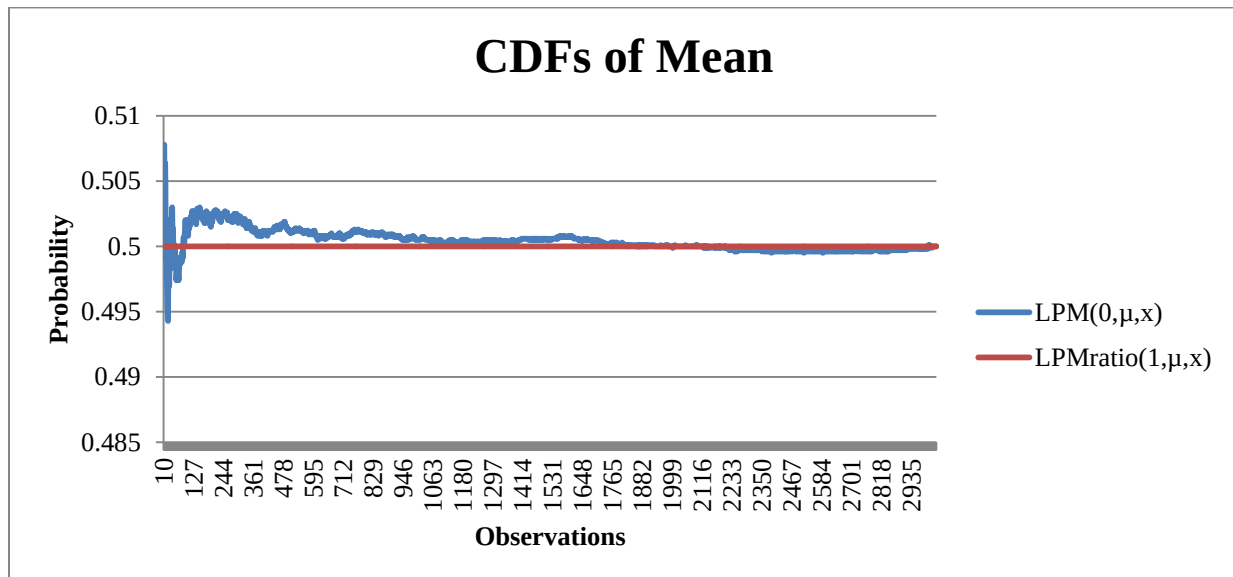


Figure 7. Differences in discrete $LPM(0,\mu,X)$ and continuous $LPMratio(1,\mu,x)$ CDFs converge when using the mean target for the Normal distribution. $LPM(0,\mu,X) \neq LPMratio(1,\mu,X)$.

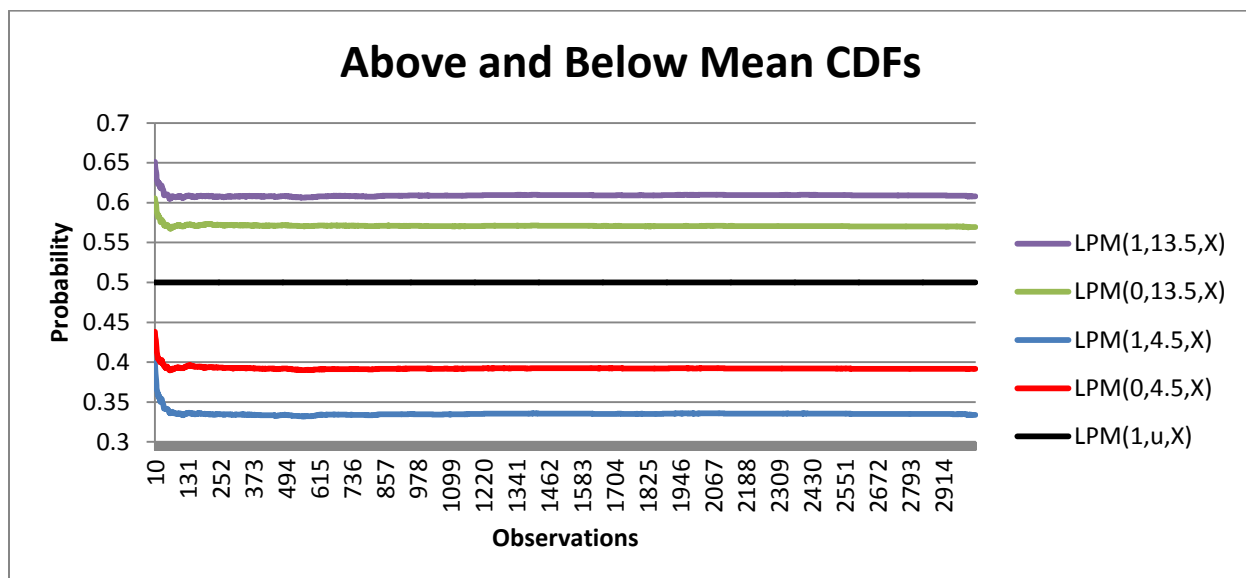


Figure 8. Different locations of the target versus the mean and relationships between discrete and continuous CDFs.

In Figure 8, we used different targets of 4.5%, 9% (mean), and 13.5% and we see that the continuous is outside of the range of the discrete measures. Note that with the mean as the target, the continuous measure is rock solid on the 50% probability.

Normal Distribution Probabilities - 5 Million Draws 300 Iteration Seeds		
Norm Prob($X \leq 0.00$) = .3085	LPM(0, 0, X) = .3085	LPM(1, 0, X) = .2208
Norm Prob($X \leq 4.50$) = .3917	LPM(0, 4.5, X) = .3917	LPM(1, 4.5, X) = .3339
Norm Prob($X \leq \text{Mean}$) = .5	LPM(0, μ , X) = .5	LPM(1, μ , X) = .5
Norm Prob($X \leq 13.5$) = .5694	LPM(0, 13.5, X) = .5694	LPM(1, 13.5, X) = .608

Table 1. Final probability estimates with 5 million observations and 300 iteration seeds averaged for the Normal distribution.

In Table 1, we see that the LPM degree 0 provides equivalent probabilities as the Normal Probability function from the IMSL library. The continuous probability using the LPM degree 1 is at 0.5 for the mean as a target and has a lower probability below the mean and a higher probability above the mean as we have noted previously.

D. Uniform Distribution

We compare our metric to the traditional uniform CDF for values less than or equal to x .

$$F(x|A, B) = \begin{cases} 0, & \text{if } x < A \\ \frac{x - A}{B - A}, & \text{if } A \leq x \leq B \\ 1, & \text{if } x > B \end{cases}$$

Table 2 below shows the convergence of our metric to the traditional method for the uniform CDF (UNDF) with a mean of 10. The results are the same as we noted for the normal distribution in Table 1.

Uniform Distribution Probabilities - 5 Million Draws 300 Iteration Seeds		
UNDF($X \leq 0.00$) = .4	LPM(0, 0, X) = .4	LPM(1, 0, X) = .3077
UNDF($X \leq 4.50$) = .445	LPM(0, 4.5, X) = .445	LPM(1, 4.5, X) = .3913
UNDF($X \leq \text{Mean}$) = .5	LPM(0, μ , X) = .5	LPM(1, μ , X) = .5
UNDF($X \leq 13.5$) = .535	LPM(0, 13.5, X) = .535	LPM(1, 13.5, X) = .5697

Table 2. Uniform distribution results illustrate convergence of $LPM(0, x, X)$ to UNDF and consistent relationship between $LPM(0, x, X)$ and $LPM_{ratio}(1, x, X)$ above and below the mean target.

E. Poisson Distribution

We compare our metric to the traditional Poisson CDF (POIDF) for values less than or equal to X.

$$f(x) = e^{-\theta} \frac{\theta^x}{x!}$$

Poisson Distribution Probabilities - 5 Million Draws 300 Iteration Seeds		
POIDF($X \leq 0.00$) = .00005	LPM(0, 0, X) = 0	LPM(1, 0, X) = 0
POIDF($X \leq 4.50$) = .0293	LPM(0, 4.5, X) = .0293	LPM(1, 4.5, X) = .0051
POIDF($X \leq \text{Mean}$) = .5151	LPM(0, μ , X) = .5151	LPM(1, μ , X) = .5
POIDF($X \leq 13.5$) = .8645	LPM(0, 13.5, X) = .8645	LPM(1, 13.5, X) = .9365

Table 3. Poisson distribution results illustrate convergence of $LPM(0, x, X)$ to POIDF and consistent relationship between $LPM(0, x, X)$ and $LPM_{ratio}(1, x, X)$ above and below the mean target.

F. Chi-Square Distribution

We compare our metric to traditional chi-square CDF (CHIDF) for values less than or equal to X .

$$F(x) = \frac{1}{2^{\frac{v}{2}} \Gamma(\frac{v}{2})} \int_0^x e^{-\frac{t}{2}} t^{\frac{v}{2}-1} dt$$

We set the degrees of freedom for the chi-square equal to one. The reason for this arbitrary selection is the distinct curve generated by this parameter value, and its likeness to the power law distribution. There is no a priori argument that the degrees of freedom will affect our methodology given its non-parametric derivation.

Chi-Squared Distribution Probabilities - 5 Million Draws 300 Iteration Seeds		
CHIDF($X \leq 0$) = 0	LPM(0, 0, X) = 0	LPM(1, 0, X) = 0
CHIDF($X \leq 0.5$) = .5205	LPM(0, 0.5, X) = .5205	LPM(1, 0.5, X) = .2087
CHIDF($X \leq 1$) = .6827	LPM(0, 1, X) = .6827	LPM(1, 1, X) = .5
CHIDF($X \leq 5$) = .9747	LPM(0, 5, X) = .9747	LPM(1, 5, X) = .989

Table 4. Chi-Square distribution results illustrate convergence of $LPM(0, x, X)$ to UNDF and consistent relationship between $LPM(0, x, X)$ and $LPM_{ratio}(1, x, X)$ above and below the mean target.

G. Continuous Distributions:

In a discrete measurement with a zero target, there is no difference between a 40% observation and a 70% observation as both will yield a single positive count in the frequency (both were observed in our normal distribution generation with $\mu=10$ and $\sigma=20$ parameter constraints). However, there is considerable area between these two observations that merely gets binned in a probability analysis. This undesirable construct also has the ubiquitous quality of scale invariance. Equation (14) measures this neglected area with its inherent variance consideration simultaneously factored with the discrete frequency analysis.

“All actual sample spaces are discrete, and all observable random variables have discrete distributions. The continuous distribution is a mathematical construction, suitable for mathematical treatment, but not practically observable.” E.J.G. Pitman (1979).

LPM_{ratio} degree of 1 ($n=q=1$) permits us to calculate the area “between the bins.” For example, in a roll of a die, the area of the function between 3.1 and 3.9 will be static for the discrete method (based on integer bins 1-6). If the distribution were actually continuous, the variance influence in LPM_{ratio} degree 1 generates an accurate measurement of the area 3.1 through 3.9 for this area between the bins - for uniform and all other distributions. Furthermore, the mean for a die roll is approximately 3.5. LPM_{ratio} degree 1 generates a 0.5 result for the CDF with the 3.5 mean as the target in a uniform distribution ranging from 1 to 6. Unfortunately, per Pitman’s observation, we are not able to generate a continuous distribution to observe and verify this notion for target values other than the mean (which we prove always equal 0.5) or endpoints (0 or 1 for sample minimum and maximum). The consistent observed relationship we demonstrated between $LPM_{ratio}(1, x, X)$ and $LPM(0, x, X)$ for targets above and below the mean, offers considerable support of the continuous estimates.

A better example to distinguish between discrete and continuous analysis is the chi-square distribution with degrees of freedom set to one. The range of the observations extended to $X=35.1$ and resembles the power law function. Considering $\mu=1.0$ and $\sigma=1.414$, the discrete probability of a mean return was 0.6827 as shown in Table 4. However, if one envisions the decreasing thin slice of area under the function all the way down the x-axis to the observation $X=35.1$, this extended result only generates a reading of one in its probability calculation of $x > \mu$. No different than an observation of $X=11$ which is also a positive count in this example. The frequency of $X=11$ is the distinguishing characteristic. This difference in area between 11 and

35.1 is considerable and is completely disregarded under discrete frequency analysis. When the variance of that deviation is considered to account for the infinite possible outcomes for the continuous variable, the probability of a mean return drops significantly to 0.5 from 0.6827.

The reason for this is straightforward, $LPM(0, x, X)$ converges to the frequency / counting data set while $LPM_{ratio}(1, x, X)$ retains its area property.

III. Joint Distribution Co-Partial Moments and UPM/LPM Analysis

In this section, we introduce the framework for the joint distribution using partial moments. For more background, Appendix B and Appendix C provide more information on joint probabilities and conditional CDFs. We also replicate the covariance matrix of a two variable normal distribution and its cosemivariance matrix with the variables' aggregated partial moment components. This information provides a toolbox that yields a positive definite symmetrical co-partial moment matrix capable of handling any target and resulting asymmetry, providing a distinct advantage over its cosemivariance counterpart.

The issue in this area traces back to the Markowitz (1959) chapter on semivariance analysis. The cosemivariance matrix in Markowitz is an endogenous matrix that is computed after the portfolio returns have been computed. Because we have to know the portfolio allocations before we can compute the portfolio returns, the cosemivariance matrix is not known until after we have solved the problem. Attempts to solve the mean-semivariance problem with an exogenous matrix, a matrix computed from the security return data, have had problems because the cosemivariance matrix is asymmetric, and therefore, not positive semi-definite. Grootveld and

Hallerbach (1999) noted that the endogenous and exogenous matrices are not equivalent. Estrada (2008), however, demonstrates that a symmetric exogenous matrix is a very good approximation for the endogenous matrix. Our purpose is to demonstrate a method that provides a positive semi-definite matrix system that preserves any asymmetry in the underlying process.

First, the LPM and the CLPM are defined as follows:

$$LPM(n, h, x) = \frac{1}{T} [\sum_{t=1}^T \max\{0, h - x_t\}^n] \quad (18)$$

$$CLPM(n, h, x|y) = \frac{1}{T} \left[\sum_{t=1}^T (\max\{0, h - x_t\}^n \cdot \max\{0, h - y_t\}^n) \right] \quad (19)$$

The Degree 1 Co-LPM (CLPM) matrix is:

$$\begin{bmatrix} LPM(2, h, x) & CLPM(1, h, y|x) \\ CLPM(1, h, x|y) & LPM(2, h, y) \end{bmatrix}$$

$$LPM(2, h, x) = CLPM(1, h, x|x) \quad (20)$$

Variance is the squared deviation

$$\sigma_x^2 = E[(x_t - \mu_{xt})^2] = \left(\frac{1}{T}\right) \cdot \sum_{t=1}^T (x_t - \mu_{xt})^2 \quad (21)$$

And it is also the deviation times itself...the covariance of itself.

$$\sigma_{xx} = \left(\frac{1}{T}\right) \cdot \sum_{t=1}^T (x_t - \mu_{xt})(x_t - \mu_{xt}) \quad (22)$$

And the covariance between 2 variables is simply

$$\sigma_{xy} = \left(\frac{1}{T}\right) \cdot \sum_{t=1}^T (x_t - \mu_{xt})(y_t - \mu_{yt}) \quad (23)$$

If semivariance from benchmark B is

$$\Sigma_{xB}^2 = E\{[\min(x - B, 0)]^2\} = \left(\frac{1}{T}\right) \cdot \sum_{t=1}^T [\min(x_t - B, 0)]^2 \quad (24)$$

Then it is also the cosemivariance of itself

$$\Sigma_{xxB} = \left(\frac{1}{T}\right) \cdot \sum_{t=1}^T [\min(x_t - B, 0)][\min(x_t - B, 0)] \quad (25)$$

And the cosemivariance between 2 variables is

$$\Sigma_{xyB} = \left(\frac{1}{T}\right) \cdot \sum_{t=1}^T [\min(x_t - B, 0)][\min(y_t - B, 0)] \quad (26)$$

If LPM degree 2 is equal to semivariance, $LPM(2, B, x) = \Sigma_{xB}^2$

$$LPM(2, B, x) = \left(\frac{1}{T}\right) \cdot \sum_{t=1}^T [\max(B - x_t, 0)]^2 \quad (27)$$

Also equals the Co-LPM degree 1 of the same variable

$$\begin{aligned} CLPM(1, B, x|x) &= \Sigma_{xB}^2 = \Sigma_{xxB} \\ &= \left(\frac{1}{T}\right) \cdot \sum_{t=1}^T [\max(B - x_t, 0)][\max(B - x_t, 0)] \end{aligned} \quad (28)$$

And the Co-LPM degree 1 between 2 variables is

$$CLPM(1, B, x|y) = \left(\frac{1}{T}\right) \cdot \sum_{t=1}^T [\max(B - x_t, 0)][\max(B - y_t, 0)] \quad (29)$$

For two symmetrical distributions x, y with $h = \mu$

$$\begin{array}{ccc} \text{Co-LPM Matrix} & = & \text{Co-UPM Matrix} \\ \left[\begin{array}{cc} LPM(2, \mu, x) & CLPM(1, \mu, y|x) \\ CLPM(1, \mu, x|y) & LPM(2, \mu, y) \end{array} \right] & = & \left[\begin{array}{cc} UPM(2, \mu, x) & CUPM(1, \mu, y|x) \\ CUPM(1, \mu, x|y) & UPM(2, \mu, y) \end{array} \right] \end{array}$$

Furthermore, the addition of the Co-LPM matrix, the Co-UPM matrix is equivalent to the covariance matrix on the main diagonal.

$$\begin{aligned} & CLPM(1, h, x|y) + CUPM(1, h, x|y) \\ &= \left[\begin{array}{cc} LPM(2, h, x) & CLPM(1, h, y|x) \\ CLPM(1, h, x|y) & LPM(2, h, y) \end{array} \right] + \left[\begin{array}{cc} UPM(2, h, x) & CUPM(1, h, y|x) \\ CUPM(1, h, x|y) & UPM(2, h, y) \end{array} \right] \\ &= \left[\begin{array}{cc} LPM(2, h, x) + UPM(2, h, x) & CLPM(1, h, y|x) + CUPM(1, h, y|x) \\ CLPM(1, h, x|y) + CUPM(1, h, x|y) & LPM(2, h, y) + UPM(2, h, y) \end{array} \right] \end{aligned}$$

The main diagonal of the aggregated matrix will retain the covariance equivalence under any asymmetry with the following relationship for all targets,

$$\sigma_x^2 = LPM(2, \mu, x) + UPM(2, \mu, x) \quad (30)$$

$$\begin{aligned} & CLPM(1, \mu, y|x) + CUPM(1, \mu, y|x) \\ &= \frac{1}{T} \left[\sum_{t=1}^T (\max\{0, \mu - x_t\} \cdot \max\{0, \mu - y_t\}) + (\max\{0, x_t - \mu\} \cdot \max\{0, y_t - \mu\}) \right] \end{aligned} \quad (31)$$

Equation (31) will generate a zero instead of a negative covariance result, ensuring a positive matrix. This zero (instead of the negative) result does not affect the preservation of information for the instances whereby one variable is above the target and one below. The addition of this observation to the complement set lowers both the LPM and UPM. In essence, nothing is still something. In R, using the ‘NNS’ package, we can verify the variance/covariance equivalence.

```
> set.seed(123); x=rnorm(100); y=rnorm(100)
> var(x)
[1] 0.8332328

#Sample:
> UPM(2,mean(x),x)+LPM(2,mean(x),x)
[1] 0.8249005

#Population:
> (UPM(2,mean(x),x)+LPM(2,mean(x),x))*(length(x)/(length(x)-1))
[1] 0.8332328

#Variance is also the co-variance of itself:
> (Co.LPM(1,1,x,x)+Co.UPM(1,1,x,x)-D.LPM(1,1,x,x)-
D.UPM(1,1,x,x))*(length(x)/(length(x)-1))
[1] 0.8332328

> cov(x,y)
[1] -0.04372107

> (Co.LPM(1,1,x,y)+Co.UPM(1,1,x,y)-D.LPM(1,1,x,y)-
D.UPM(1,1,x,y))*(length(x)/(length(x)-1))
[1] -0.04372107
```


A. Complement Set Matrix

To further analyze the information in the $(CLPM + CUPM)^C$ complement set from diverging target returns between variables, we introduce two new metrics - the diverging lower partial moment ($DLPM$) and diverging upper partial moment ($DUPM$).

$$DLPM(q|n, h, x|y) = \frac{1}{T} \left[\sum_{t=1}^T (\max\{x_t - h, 0\}^q \cdot \max\{0, h - y_t\}^n) \right] \quad (32)$$

$$DUPM(n|q, h, x|y) = \frac{1}{T} \left[\sum_{t=1}^T (\max\{0, h - x_t\}^n \cdot \max\{y_t - h, 0\}^q) \right] \quad (33)$$

Equation (32) provides the divergent LPM for variable Y given a positive target deviation for variable X from shared target h , with the LPM and UPM degrees (n and q respectively) explained earlier in equations 1 and 2. For example, given a 20% observation for variable X and a shared target of 0%, a -10% observation for variable Y will generate a larger $DLPM$ than a -5% observation for variable Y .

Conversely, equation (33) provides the divergent UPM for variable Y given a negative target deviation for variable X .

The matrix of each divergent partial moment will be aggregated to represent the divergent partial moment matrix (DPM). One key feature of this matrix is the main diagonal consists of all zeros since the divergent partial moment of the same variable does not exist. The degree 1 DPM is presented below.

$$\begin{bmatrix} 0 & DPM(1|1, h, x|y) \\ DPM(1|1, h, y|x) & 0 \end{bmatrix} = \begin{bmatrix} 0 & DLPM(1|1, h, x|y) \\ DLPM(1|1, h, y|x) & 0 \end{bmatrix} + \begin{bmatrix} 0 & DUPM(1|1, h, x|y) \\ DUPM(1|1, h, y|x) & 0 \end{bmatrix} \quad (34)$$

Since there only exists four possible interactions between two variables,

$X \leq \text{target}, Y \leq \text{target}$	$CLPM(n, h, x y)$
$X \leq \text{target}, Y > \text{target}$	$DUPM(n q, h, x y)$
$X > \text{target}, Y \leq \text{target}$	$DLPM(q n, h, x y)$
$X > \text{target}, Y > \text{target}$	$CUPM(q, l, x y)$

we can clearly see that the sum of the degree 0 probability matrices of all four interactions must equal one, explaining the entire multivariate distribution.

The distinct advantage for the partial moments over semivariance as the preferred below target analysis method, is the ability for the partial moments to compensate for any asymmetry.

Under symmetry,

Cosemivariance Matrix = $\frac{1}{2}$ Covariance Matrix

$$\begin{bmatrix} \Sigma_{xx\mu} & \Sigma_{xy\mu} \\ \Sigma_{yx\mu} & \Sigma_{yy\mu} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{bmatrix}$$

$$\begin{bmatrix} \Sigma_{xx\mu} & \Sigma_{xy\mu} \\ \Sigma_{yx\mu} & \Sigma_{yy\mu} \end{bmatrix} + \begin{bmatrix} \Sigma_{xx\mu} & \Sigma_{xy\mu} \\ \Sigma_{yx\mu} & \Sigma_{yy\mu} \end{bmatrix} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{bmatrix} \quad (35)$$

Minimizing $\begin{bmatrix} \Sigma_{xx\mu} & \Sigma_{xy\mu} \\ \Sigma_{yx\mu} & \Sigma_{yy\mu} \end{bmatrix}$ creates an imbalance that has no offsetting components to equal the covariance matrix when added to itself. The minimizing of the LPM matrix and the DLPM matrix has a simultaneous inverse effect of increasing the UPM matrix and DUPM matrix, ergo compensating for any asymmetry. This balancing effect holds for any target, not just μ .

$$\begin{bmatrix} LPM(2, \mu, x) & CLPM(1, \mu, x|y) \\ CLPM(1, \mu, y|x) & LPM(2, \mu, y) \end{bmatrix} - \begin{bmatrix} 0 & DPM(1|1, \mu, x|y) \\ DPM(1|1, \mu, y|x) & 0 \end{bmatrix} + \begin{bmatrix} UPM(2, \mu, x) & CUPM(1, \mu, x|y) \\ CUPM(1, \mu, y|x) & UPM(2, \mu, y) \end{bmatrix} = \begin{bmatrix} \sigma_{xx} & \sigma_{yx} \\ \sigma_{xy} & \sigma_{yy} \end{bmatrix} \quad (36)$$

Each of the co-partial moment matrices is positive symmetrical semi-definite, ensuring a positive symmetrical definite aggregate matrix, thus avoiding the endogenous/exogenous matrix problem described by Grootveld and Hallerbach (1999) and Estrada (2008).

IV. Conclusions

We have demonstrated how the *LPM* degree 0 is equal to the traditionally derived CDF of any assumed distribution. $LPM(0, x, X)$ converges to:

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt,$$

$$F(x|A, B) = \begin{cases} 0, & \text{if } x < A \\ \frac{x-A}{B-A}, & \text{if } A \leq x \leq B, \\ 1, & \text{if } x > B \end{cases}$$

$$f(x) = e^{-\theta} \frac{\theta^x}{x!},$$

$$F(x) = \frac{1}{2^{\frac{v}{2}} \Gamma(\frac{v}{2})} \int_0^x e^{-\frac{t}{2}} t^{\frac{v}{2}-1} dt.$$

The obvious benefit is the distribution agnostic manner of this direct computation, which consumes far less time and cpu effort than bootstrapping a discrete estimate. Furthermore, the stability of the partial moments versus each of the distribution estimates is yet another benefit of our method. Finally, the ability to derive results for a truly continuous variable emphasizes the flexibility of this method.

Any computer generated sample and analysis thereof, is that of a discrete variable. A histogram and bins as commonly performed in Excel by practitioners and academics alike ignores a large area under the function due to this discrete classification. The addition of bins with increased observations does not fill in the area and converge to the continuous area estimate; it merely creates larger quantities of smaller areas thus keeping the total area constant. Equation (14) makes no such concessions and generates the theoretical continuous area, while maintaining the relationship identified in Equation (15). We note how the continuous CDF is

much more pronounced the further from the mean the integral is - compensating for the asymmetry of the additional area “between the bins” that is placed in the proceeding bin during discrete analysis.

Benoit Mandelbrot notes the shorter the measuring instrument, the larger the coastline of Britain; ultimately yielding a result of infinity. This line of reasoning is commensurate with the continuous CDF versus its discrete counterpart; and the infinitesimal subintervals of a continuous distribution. We hope that further research on this method and its applications eventually finds its way to various fields of study.

We show that the Cumulative Distribution Function (CDF) is represented by the ratio of the lower partial moment ratio (LPM_{ratio}) to the distribution for the interval in question. The addition of the upper partial moment ratio (UPM_{ratio}) enables us to create probability density functions (PDF) for any function or distribution without prior knowledge of its characteristics. The ability to derive the CDF and PDF without any distributional assumptions yields a more accurate calculation devoid of any error terms present from a less than perfect goodness of fit, as well as critical information about the tails of the distribution. This foundation is then used to develop conditional probabilities and joint distribution co-partial moments. The resulting toolbox allows us to propose a new formulation for UPM/LPM analysis.

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Appendix A:

In this section we address any sample size concerns the reader may logically infer. Since these concerns are not specific to our methodology but rather to statistics in general, we offer the results of a separate study comparing the deviations from the large sample sizes reported in the main body of this paper.

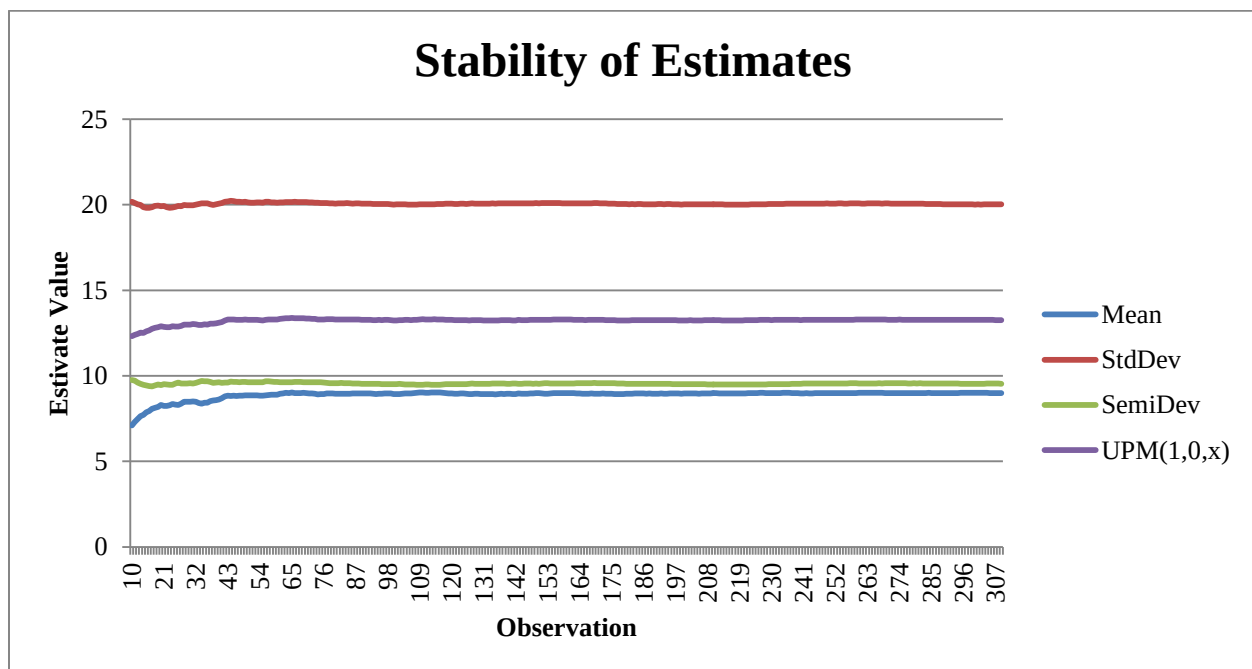


Figure 1a. Visual representation of the stabilization of statistics as sample size increases.

Test	Mean	StdDev	SemiDev	Variance	LPM (0,0,x)	LPM (1,0,x)	UPM (0,0,x)	UPM (1,0,x)	LPM (0,μ,x)	LPM (1,4.5,x)	LPM (1,13.5,x)	LPM (0,4.5,x)	LPM (0,13.5,x)	Norm Prob (4.5)	Norm Prob (13.5)	Norm Prob (0)
MAD 30-100	0.1258	0.0479	0.0409	2.2496	0.0024	0.0582	0.0024	0.0774	0.0012	0.0043	0.0031	0.0023	0.0019	0.0025	0.0024	0.0024
MAD 30-300	0.0678	0.0378	0.0395	2.2107	0.0014	0.0506	0.0014	0.037	0.0013	0.0026	0.0015	0.0014	0.0012	0.0014	0.0012	0.0014
MAD 30-3000	0.021	0.0132	0.0266	0.7485	0.0005	0.0125	0.0005	0.0121	0.0006	0.0007	0.0007	0.0005	0.0004	0.0004	0.0004	0.0004
MAD 100-5 Million	0.0194	0.0106	0.026	0.4914	0.0005	0.011	0.0005	0.0107	0.0006	0.0006	0.0006	0.0004	0.0004	0.0004	0.0004	0.0004
MAD 3000-5 Million	0.0039	0.0043	0.0031	0.1833	0.0001	0.0021	0.0001	0.0027	0	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001

Table 1a. Final Stability of Estimates Test for the Normal distribution.

Test	Mean	StdDev	SemiDev	Variance	LPM (0,0,x)	LPM (1,0,x)	UPM (0,0,x)	UPM (1,0,x)	LPM (0,μ,x)	LPM (1,4.5,x)	LPM (1,13.5,x)	LPM (0,4.5,x)	LPM (0,13.5,x)
MAD 30-100	0.031	0.0208	0.0280	1.9174	0.0005	0.0163	0.0005	0.0206	0.0005	0.0008	0.0006	0.0006	0.0005
MAD 30-300	0.056	0.0184	0.0516	1.9271	0.0008	0.0228	0.0008	0.0342	0.0009	0.001	0.0012	0.0009	0.0007
MAD 30-3000	0.0342	0.0140	0.0330	0.7431	0.0003	0.0169	0.0003	0.0193	0.0002	0.0006	0.0007	0.0003	0.0004
MAD 100-5 Million	0.0283	0.0133	0.0245	0.7108	0.0003	0.0134	0.0003	0.0173	0.0002	0.0005	0.0006	0.0002	0.0004
MAD 3000-5 Million	0.0067	0.0029	0.0023	0.1725	0.0001	0.0019	0.0001	0.0049	0	0.0001	0.0001	0.0001	0.0001

Table 2a. Final Stability of Estimates Test for the Uniform distribution.

Test	Mean	StdDev	SemiDev	Variance	LPM (0,4.5,x)	LPM (1,4.5,x)	UPM (0,4.5,x)	UPM (1,4.5,x)	LPM (0, μ ,x)	LPM (1,0,x)	LPM (1,13.5,x)	LPM (0,0,x)	LPM (0,13.5,x)
MAD 30-100	0.0439	0.0322	0.0327	0.2358	0.0053	0.0111	0.0053	0.0327	0.0027	0	0.0012	0	0.0016
MAD 30-300	0.0486	0.0353	0.0346	0.2536	0.0055	0.0109	0.0055	0.0376	0.0029	0	0.0012	0	0.0014
MAD 30-3000	0.0091	0.0115	0.0091	0.079	0.0012	0.0027	0.0012	0.0069	0.0027	0	0.0005	0	0.0004
MAD 100-5 Million	0.0056	0.0082	0.0049	0.0542	0.0006	0.0014	0.0006	0.0046	0.0027	0	0.0005	0	0.0004
MAD 3000-5 Million	0.0009	0.0038	0.0022	0.0242	0.0002	0.0004	0.0002	0.0006	0.0031	0	0.0001	0	0.0001

Table 3a. Final Stability of Estimates Test for the Poisson distribution.

Test	Mean	StdDev	SemiDev	Variance	LPM (0,0.5,x)	LPM (1,0.5,x)	UPM (0,0.5,x)	UPM (1,0.5,x)	LPM (0, μ ,x)	LPM (1,0,x)	LPM (1,5,x)	LPM (0,0,x)	LPM (0,5,x)
MAD 30-100	0.036	0.074	0.0009	0.2792	0.0033	0.0009	0.0033	0.0352	0.0032	0	0.004	0	0.0055
MAD 30-300	0.0371	0.0747	0.0012	0.2745	0.0031	0.0011	0.0031	0.036	0.0029	0	0.0039	0	0.0053
MAD 30-3000	0.0102	0.0231	0.0006	0.075	0.0014	0.0006	0.0014	0.0096	0.0008	0	0.001	0	0.0015
MAD 100-5 Million	0.0057	0.0186	0.0004	0.0554	0.0009	0.0004	0.0009	0.0053	0.0006	0	0.0007	0	0.0008
MAD 3000-5 Million	0.0027	0.0095	0.0001	0.0277	0.0003	0.0001	0.0003	0.0026	0.0002	0	0.0003	0	0.0002

Table 4a. Final Stability of Estimates Test for the Chi-Squared distribution.

Appendix B:**Conditional Probabilities:**

We illustrate how the partial moment ratios can also emulate conditional probability calculations. We re-visualize the Venn diagram areas in Figure 1b as distribution areas from which the *LPM* and *UPM* can be observed.

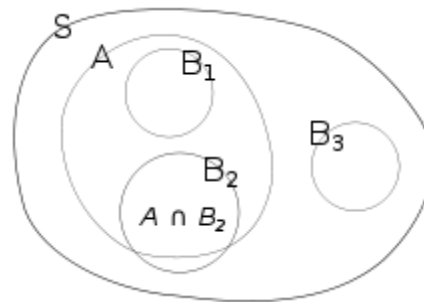


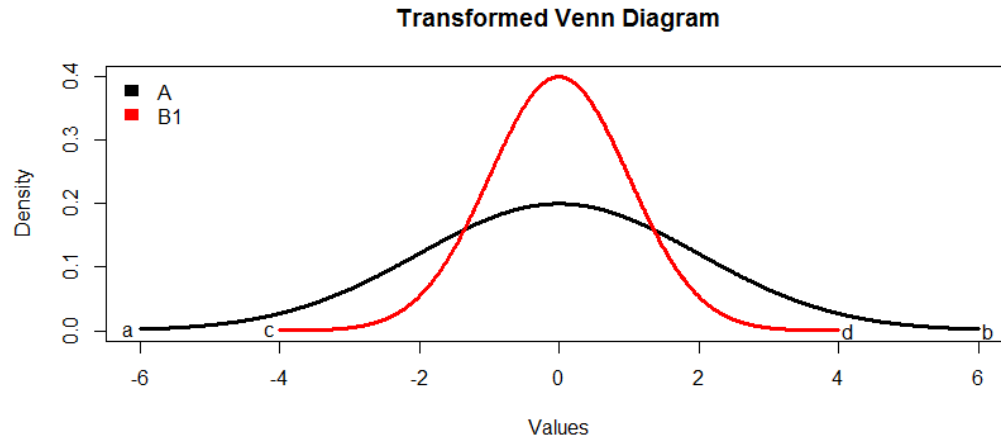
Figure 1b. Venn diagram illustrating conditional probabilities of different areas in the sample space, S.

$$P(A|B_1) = 1$$

$$P(A|B_2) \approx 0.85$$

$$P(A|B_3) = 0.$$

The conditional probability $P(A|B_1) = 1$.

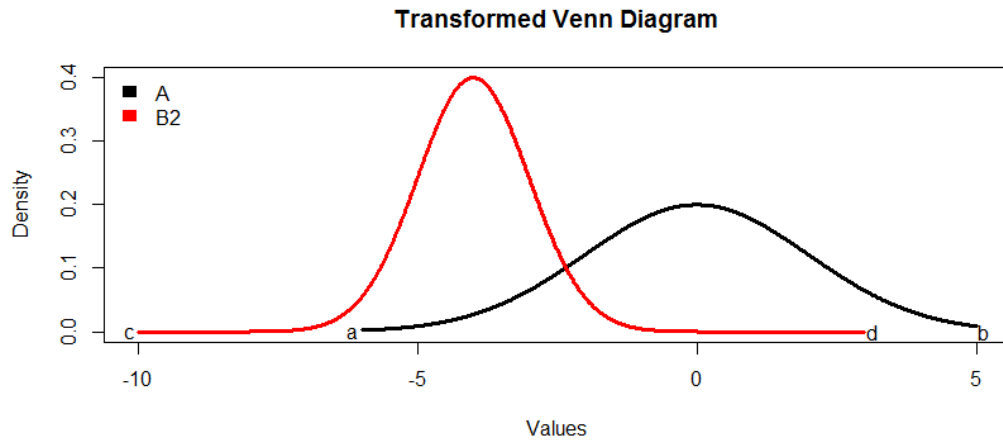


$$1 = 1 - LPM(0, a, B_1) - UPM(0, b, B_1) \quad (B.1)$$

$$1 = UPM(0, a, B_1) - UPM(0, b, B_1) \quad (B.2)$$

$$1 = (1) - (0)$$

The conditional probability $P(A|B_2) \approx 0.85$.

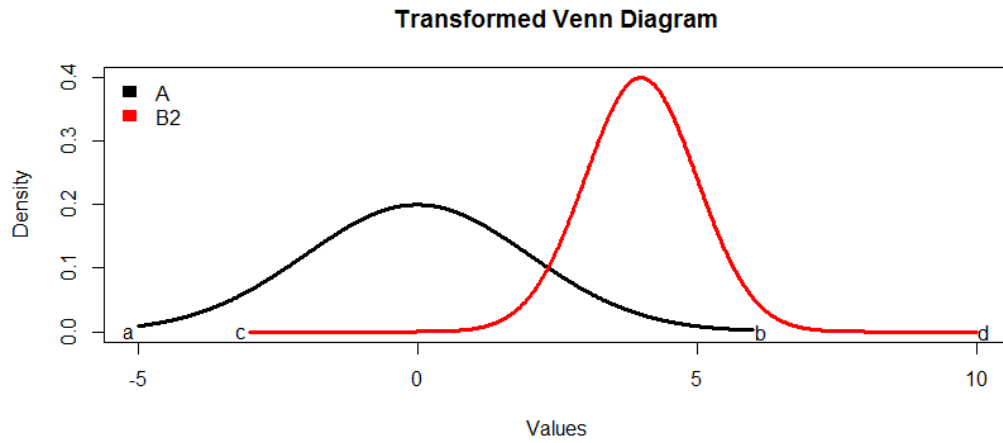


$$0.85 = 1 - LPM(0, a, B_2) - UPM(0, b, B_2) \quad (B.4)$$

$$0.85 = UPM(0, a, B_2) - UPM(0, b, B_2) \quad (B.5)$$

$$0.85 = (.85) - (0)$$

The conditional probability $P(A|B_2) \approx 0.85$.

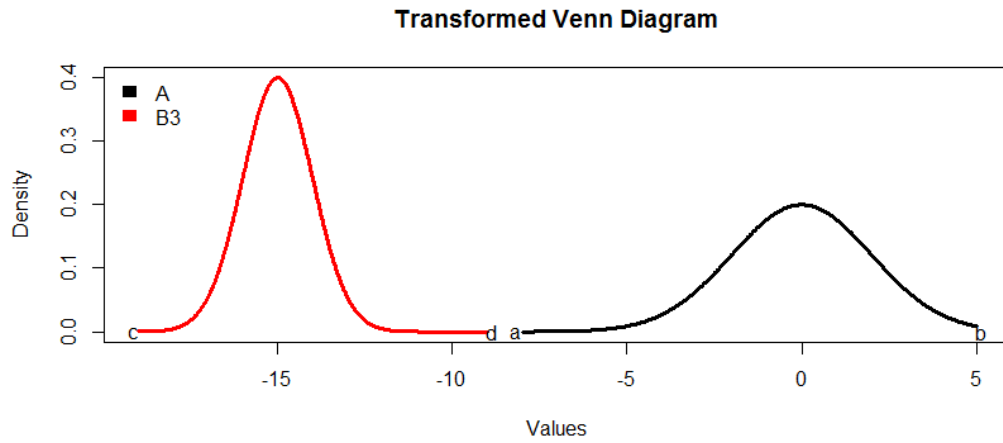


$$0.85 = 1 - LPM(0, a, B_2) - UPM(0, b, B_2) \quad (B.7)$$

$$0.85 = UPM(0, a, B_2) - UPM(0, b, B_2) \quad (B.8)$$

$$0.85 = (1) - (.15)$$

The conditional probability $P(A|B_3) = 0$.

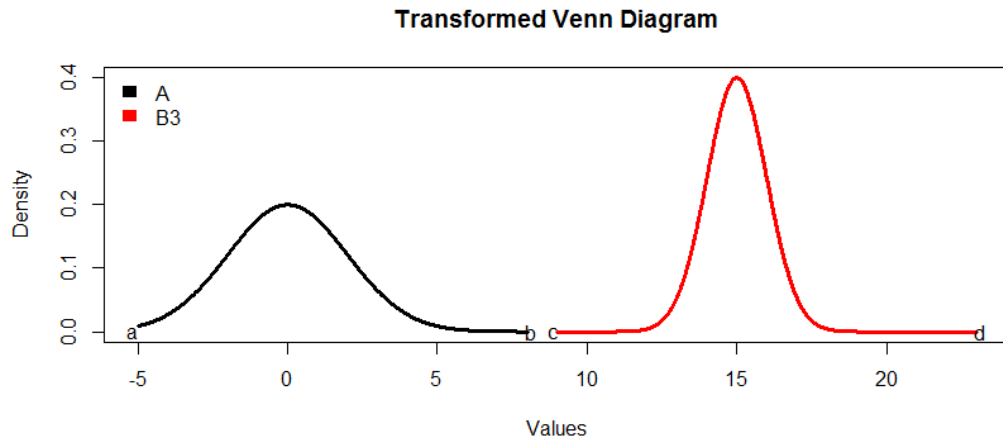


$$0 = 1 - LPM(0, a, B_3) - UPM(0, b, B_3) \quad (B.10)$$

$$0 = UPM(0, a, B_3) - UPM(0, b, B_3) \quad (B.11)$$

$$0 = (0) - (0)$$

The conditional probability $P(A|B_3) = 0$.



$$0 = 1 - LPM(0, a, B_3) - UPM(0, b, B_3) \quad (B.13)$$

$$0 = UPM(0, a, B_3) - UPM(0, b, B_3) \quad (B.14)$$

$$0 = (1) - (1)$$

Bayes' Theorem:

Bayes' theorem will also generate the conditional probability of A given B , $P(A|B)$ with the formula

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}.$$

Where the probability of A is represented by,

$$P(A) = \frac{\text{Area of } A}{\text{Area of total sample } S} = UPM(0, a, A)$$

And the probability of B is represented by,

$$P(B) = \frac{\text{Area of } B}{\text{Area of total sample } S} = UPM(0, c, B)$$

Where e is the minimum value target of area (distribution) S ; just as a and c are for areas (distributions) A and B respectively (d and b are maximum respective value targets). Thus, if the conditional probability of B given A is (per equation B.2),

$$P(B|A) = \frac{CUPM(0|0, c|a, B|A)}{UPM(0, a, A)}$$

Then,

$$P(A|B) = \frac{\frac{CUPM(0|0, c|a, B|A)}{UPM(0, a, A)} UPM(0, a, A)}{UPM(0, c, B)}$$

Cancelling out $P(A)$ leaves us with Bayes' theorem represented by partial moments, and our conditional probability on the right side of the equality.

$$P(A|B) = \frac{CUPM(0|0, c|a, B|A)}{UPM(0, c, B)}$$

The following table of the canonical breast cancer test example will help place the partial moments with their respective outcomes (R commands in red):

- 1% of women have breast cancer (and therefore 99% do not).
- 80% of mammograms detect breast cancer when it is there (and therefore 20% miss it).
- 10% of mammograms detect breast cancer when it's not there (and therefore 90% correctly return a negative result).
- Using -1 for C & TN instances, and 1 for NC & TP instances⁶

	Cancer (1%)	No Cancer (99%)	Y variable
Test Positive	Co.UPM(0,0,T,C,0,0)=.008	D.LPM(0,0,T,C,0,0)=.099	UPM(0,0,T) = .107
Test Negative	D.UPM(0,0,T,C,0,0)=.002	Co.LPM(0,0,T,C,0,0)=.891	LPM(0,0,T) =.893
X variable	UPM(0,0,C) = .01	LPM(0,0,C) = .99	UPM+LPM=1

Appendix C: Joint CDFs and UPM/LPM Correlation Analysis

Joint CDFs:

The discrete probability that both X is less than some target h_x and Y is less than some target h_y simultaneously is simply the degree 0 co-LPM provided earlier in equation (29).

$$\Pr[x \leq h_x, y \leq h_y] = CLPM(0, h_x | h_y, x | y) \quad (C. 1)$$

⁶ In R representing 1000 individuals:

> C=c(rep(1,8),rep(-1,990),rep(1,2)); T=c(rep(1,107),rep(-1,893))

This is the discrete CDF of the joint distribution, just how we prove $LPM(0, h, X)$ is the discrete CDF of the univariate distribution.

Where,

$$0 \leq CLPM(0, h_x|h_y, x|y) \leq 1 \quad (C. 2)$$

$CLPM(0, h_x|h_y, x|y)$ has the following properties for various correlations between the two variables ρ_{xy} , when $h_x = h_y$.⁷

- If $\rho_{xy} = 1$; $CLPM(0, h_x|h_y, x|y) = \min\{LPM(0, h_x, x), LPM(0, h_y, y)\}$.
- If $\rho_{xy} = 0$; $CLPM(0, h_x|h_y, x|y) = h_x \cdot h_y$
- If $\rho_{xy} = -1$; $CLPM(0, h_x|h_y, x|y) = 0$.

An example may help illustrate the relationship. Let's assume the same target $h_x = h_y$ which we arbitrarily select to the 5% CDF level for two normal distributions with $\mu = 9$ and $\sigma = 20$. We then ask, what's the probability that both variables will be in the lower 5% of their distribution simultaneously under different correlations?

⁷ We leave further asymmetric target analysis for future research.

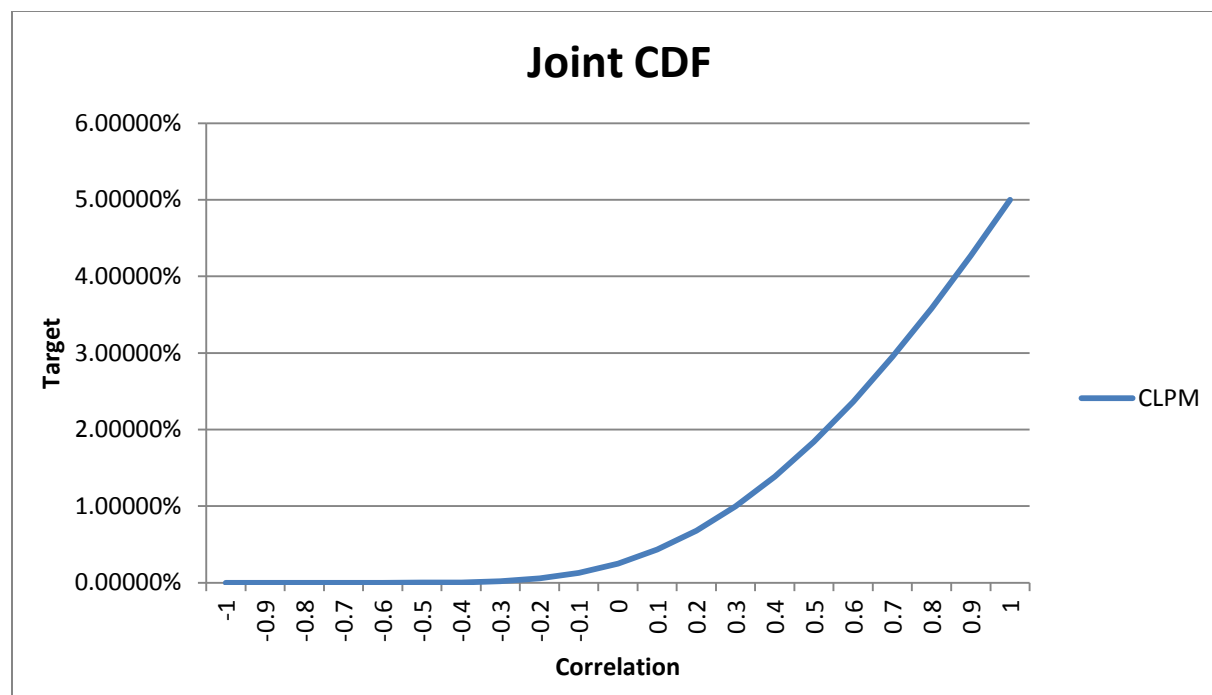


Figure 1C. Hypothetical 5% shared target on two variables (x, y) and the joint CDF for various correlations.

We can deduce the correlation between the assets only with knowledge of the *CLPM* and $h_x|h_y$. For example, with both our variables and their 5% targets, if the $CLPM(0, h_x|h_y, x|y) = 0.25\%$ we know that $\rho_{xy} = 0$.

Equation C.3 will provide the implied correlation for an observed discrete joint CDF, $CLPM(0, h_x|h_y, x|y)$. Lucas (1995) provides a framework for estimating the correlation between two events with the following equation which substitutes a binomial event into the standard Pearson correlation coefficient:

$$Corr(A, B) = \frac{P(A \text{ and } B) - P(A) \times P(B)}{[P(A)(1 - P[A])]^{\frac{1}{2}} \times [P(B)(1 - P[B])]^{\frac{1}{2}}} \quad (C.3)$$

From which we can substitute the partial moments for our events ($x < h_x, y < h_y$), yielding

$$\rho_{xy} = \frac{CLPM(0, h_x | h_y, x | y) - LPM(0, h_x, x) \cdot LPM(0, h_y, y)}{\sqrt{[LPM(0, h_x, x) \cdot UPM(0, h_x, x)] \cdot [LPM(0, h_y, y) \cdot UPM(0, h_y, y)]}} \quad (C. 4)$$

From our $h_x = h_y = 5\%$ example,

$$\rho_{xy} = \frac{0.25\% - (5\%)(5\%)}{\sqrt{[5\% \cdot 95\%] \cdot [5\% \cdot 95\%]}}$$

$$\rho_{xy} = 0.$$

If the first term in the numerator ($CLPM(0, h_x | h_y, x | y)$) equals 0.25%, the implied correlation for that joint CDF is zero. This example also illustrates the independence criterion ($h_x \cdot h_y$) from a zero correlation.

Partial Moment (Nonlinear) Correlations:

Avoiding the linear dependence of the Pearson coefficient from which Lucas' coefficient is derived, we can use the following relationship in Equation C.5 to determine the nonlinear correlation between two variables.

$$\rho_{xy} = \frac{[CLPM(0, h_x | h_y, x | y) - DLPM(0, h_x | h_y, x | y) - DUPM(0, h_x | h_y, x | y) + CUPM(0, h_x | h_y, x | y)]}{[CLPM(0, h_x | h_y, x | y) + DLPM(0, h_x | h_y, x | y) + DUPM(0, h_x | h_y, x | y) + CUPM(0, h_x | h_y, x | y)]} \quad (C. 5)$$

If there is a -1 correlation, then the returns between the variables will always be divergent, thus

$$\rho_{xy} = \frac{[0 - DLPM(0, h_x|h_y, x|y) - DUPM(0, h_x|h_y, x|y) + 0]}{[0 + DLPM(0, h_x|h_y, x|y) + DUPM(0, h_x|h_y, x|y) + 0]} = -1 \quad (C.6)$$

If there is a perfect correlation between two variables, then there will be no divergent returns, thus

$$\rho_{xy} = \frac{[CLPM(0, h_x|h_y, x|y) - 0 - 0 + CUPM(0, h_x|h_y, x|y)]}{[CLPM(0, h_x|h_y, x|y) + 0 + 0 + CUPM(0, h_x|h_y, x|y)]} = 1 \quad (C.7)$$

If there is zero correlation between two variables, then the co- and divergent returns will be of equal frequency or magnitude (degree zero and degree one respectively),

$$CLPM(0, h_x|h_y, x|y) = DLPM(0, h_x|h_y, x|y) = DUPM(0, h_x|h_y, x|y) = CUPM(0, h_x|h_y, x|y)$$

Thus,

$$\rho_{xy} =$$

$$\frac{[CLPM(0, h_x|h_y, x|y) - DLPM(0, h_x|h_y, x|y) - DUPM(0, h_x|h_y, x|y) + CUPM(0, h_x|h_y, x|y)]}{[CLPM(0, h_x|h_y, x|y) + DLPM(0, h_x|h_y, x|y) + DUPM(0, h_x|h_y, x|y) + CUPM(0, h_x|h_y, x|y)]} = 0 \quad (C.8)$$

Degree one can be substituted to generate correlations whereby the magnitude of the target deviations are compared, generating a dependence coefficient.

Continuous Joint CDF:

The continuous joint CDF can be obtained with the following equation; whereby the ratio of $CLPM(1, h_x|h_y, x|y)$ to the entire degree 1 joint distribution will generate the probability percentage. Thus,

$$CLPM_{ratio}(1, h_x|h_y, x|y) = \frac{CLPM(1, h_x|h_y, x|y)}{[LPM(1, h_x, x) \cdot LPM(1, h_y, y)] + [UPM(1, h_x, x) \cdot UPM(1, h_y, y)]} \quad (C.9)$$

$$\Pr[x \leq h_x, y \leq h_y] = CLPM_{ratio}(1, h_x|h_y, x|y) \quad (C.10)$$